

Introduction to Field Programmable Gate Arrays

Digital System Design (EC-410) Week 1, Lecture 1

Dr Muhammad Yasin

Outline

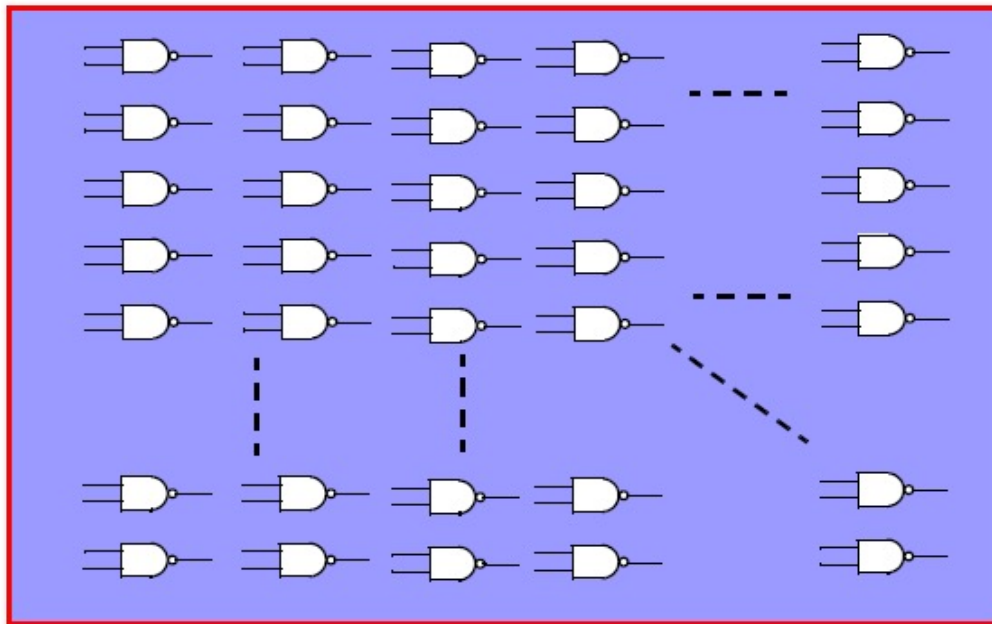
- Historical introduction.
- Basics of digital design.
- FPGA structure.

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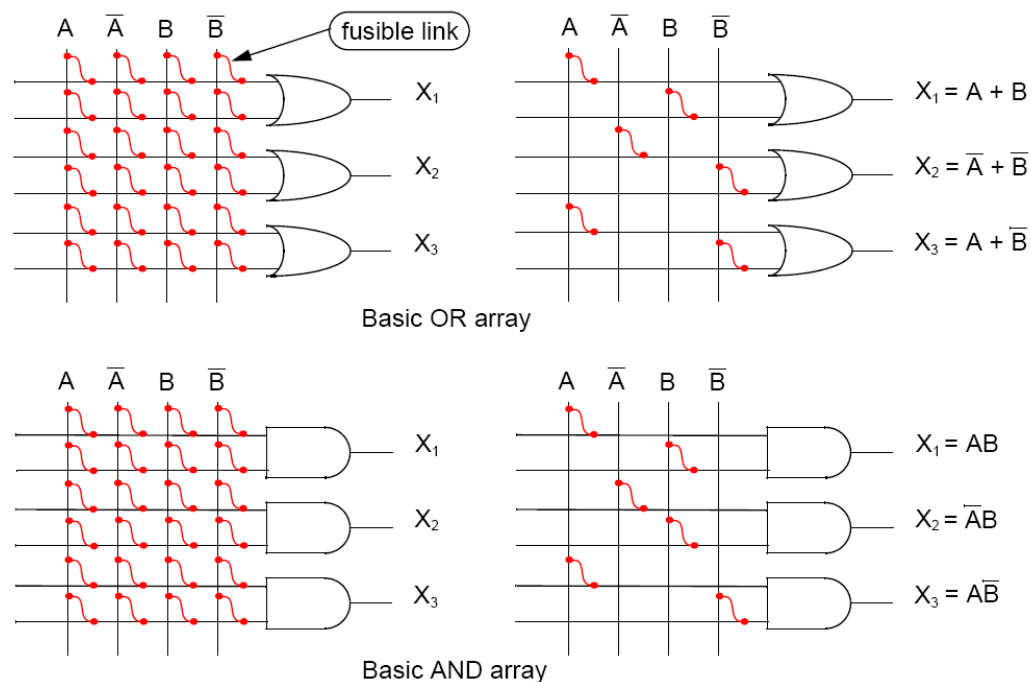
Historical Introduction

- In the beginning, digital design was done with the '74 series of chips.
- Some people would design their own chips based on Gate Arrays, which were nothing else than an array of NAND gates:



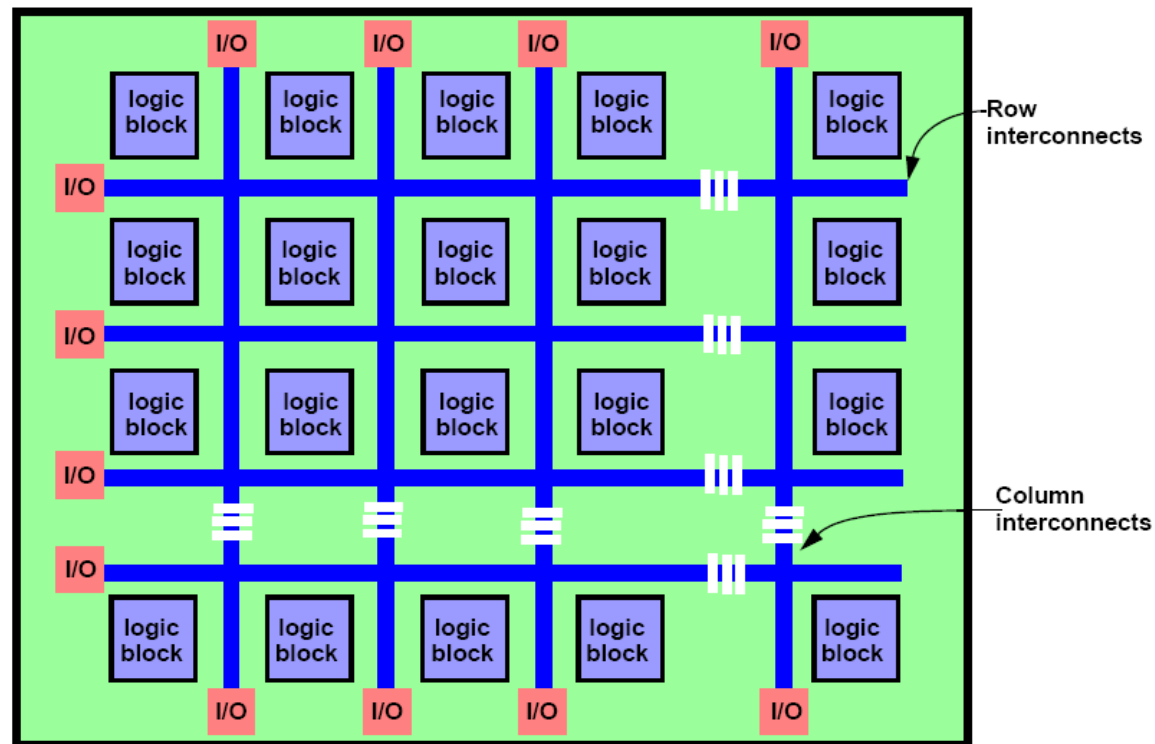
Historical Introduction

- The first programmable chips were PLAs (Programmable Logic Arrays): two level structures of AND and OR gates with user programmable connections.
- Programmable Array Logic devices were an improvement in structure and cost over PLAs. Today such devices are generically called Programmable Logic Devices (PLDs).



Historical introduction

- A complex PLD (CPLD) is nothing else than a collection of multiple PLDs and an interconnection structure.
- Compared to a CPLD, a Field Programmable Gate Array (FPGA) contains a much larger number of smaller individual blocks + large interconnection structure that dominates the entire chip.



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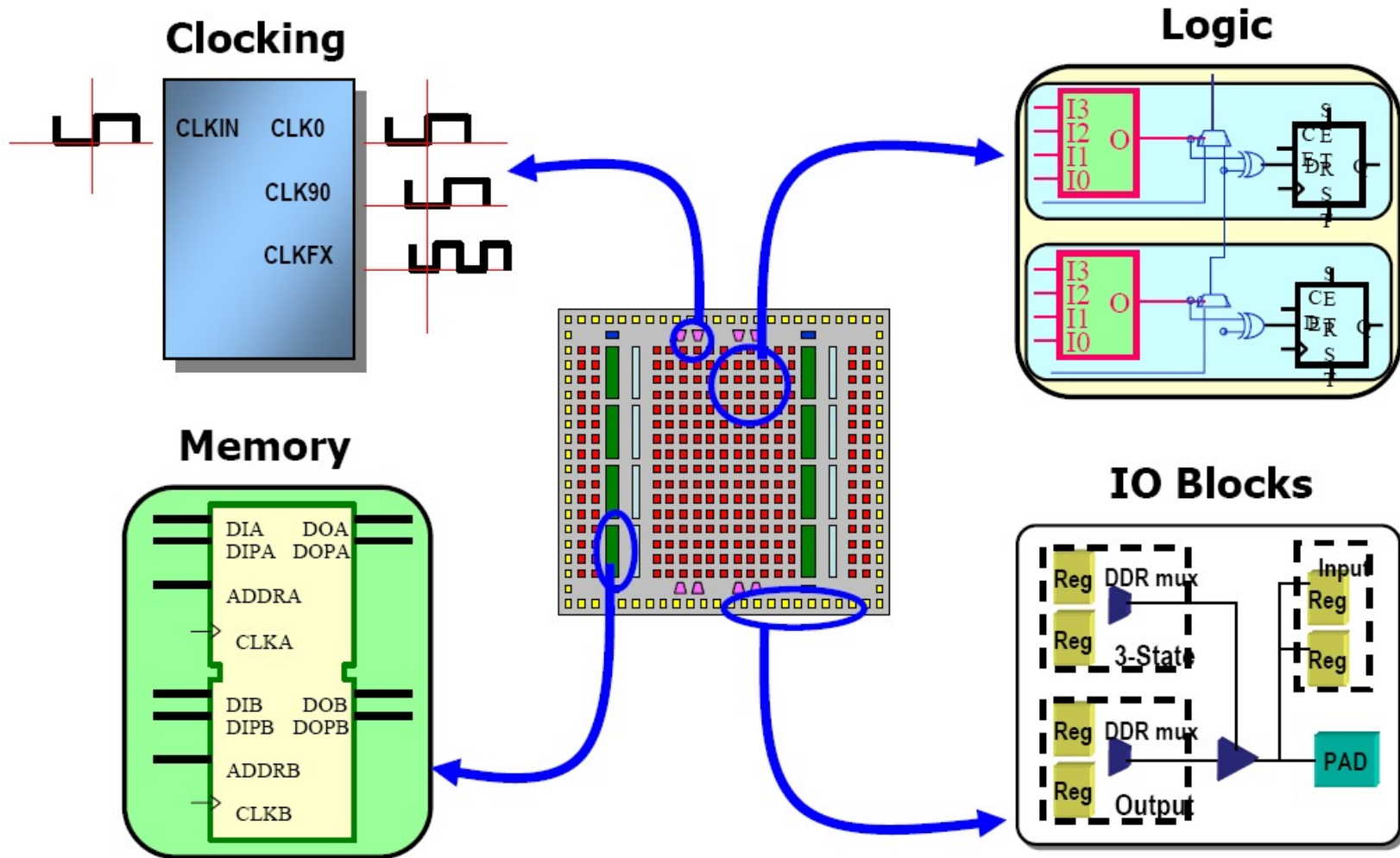
Basics of digital design

- Unless you really know what you are doing, stick to synchronous design: sandwiching bunches of combinational logic in between flip flops.
- Combinational logic: state of outputs depend on current state of inputs alone (forgetting about propagation delays for the time being). E.g. AND, OR, mux, decoder, adder...
- D-type Flip flops propagate D to Q upon a rising edge in the clk input.
- Synchronous design simplifies design analysis, which is good given today's logic densities.

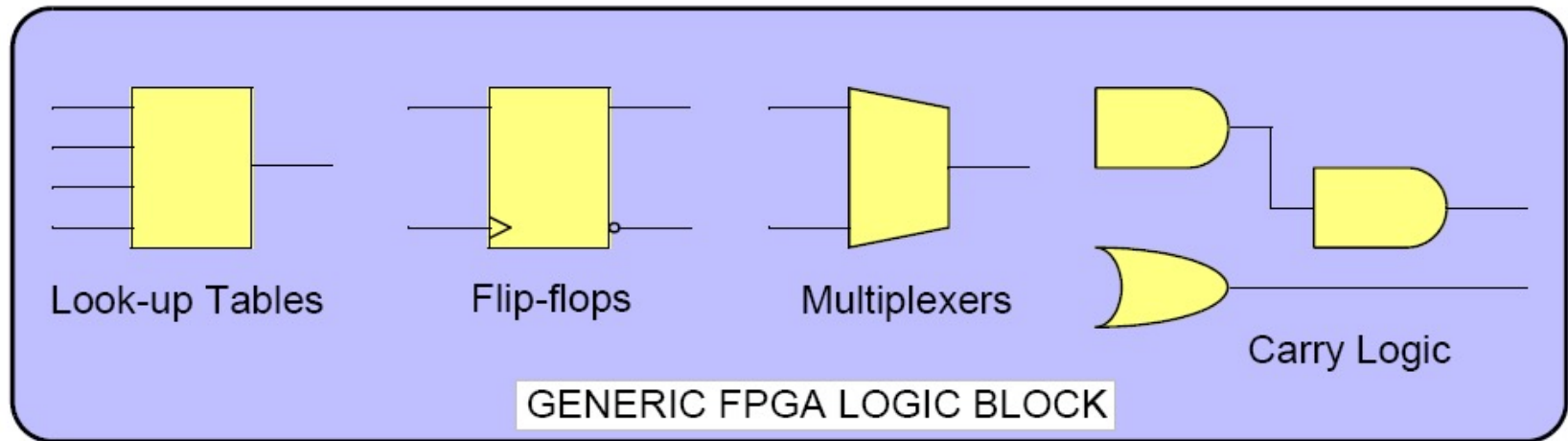
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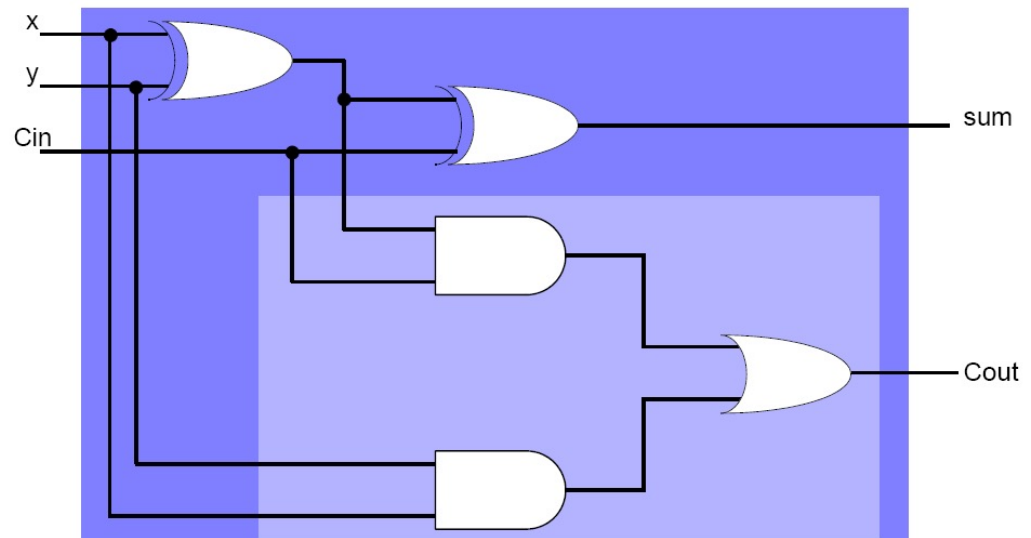
Basic FPGA architecture



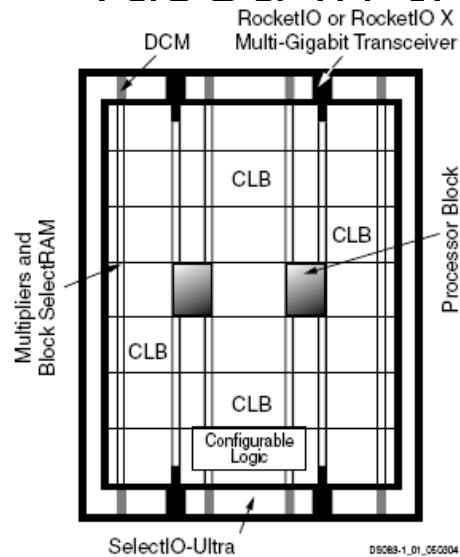
The logic block: a summary view



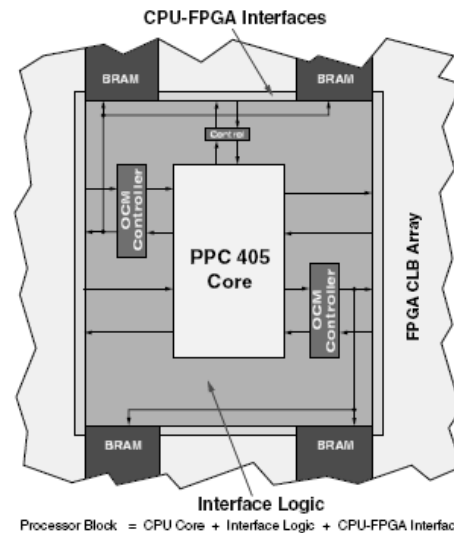
Example: using a LUT as a full adder.



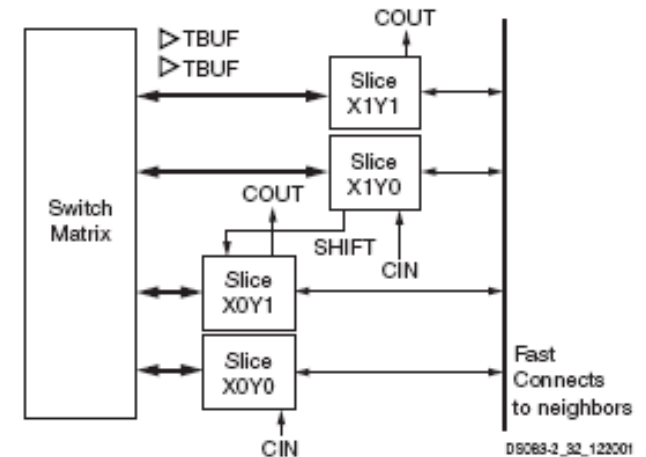
A practical example: Xilinx Virtex II Pro family (used in the lab)



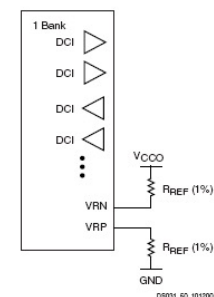
Overview



Embedded PowerPC

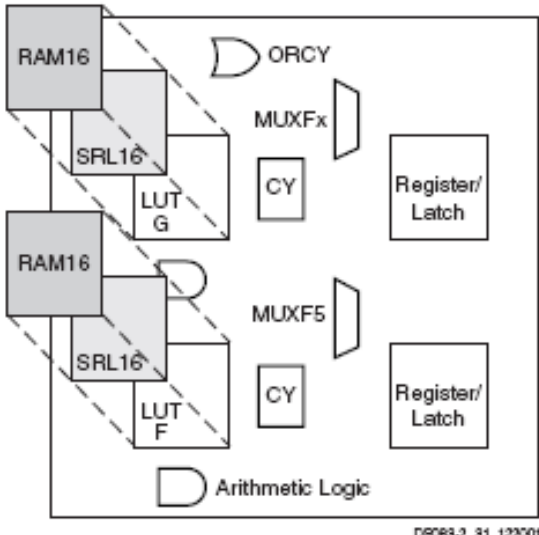


Configurable Logic Block (CLB)

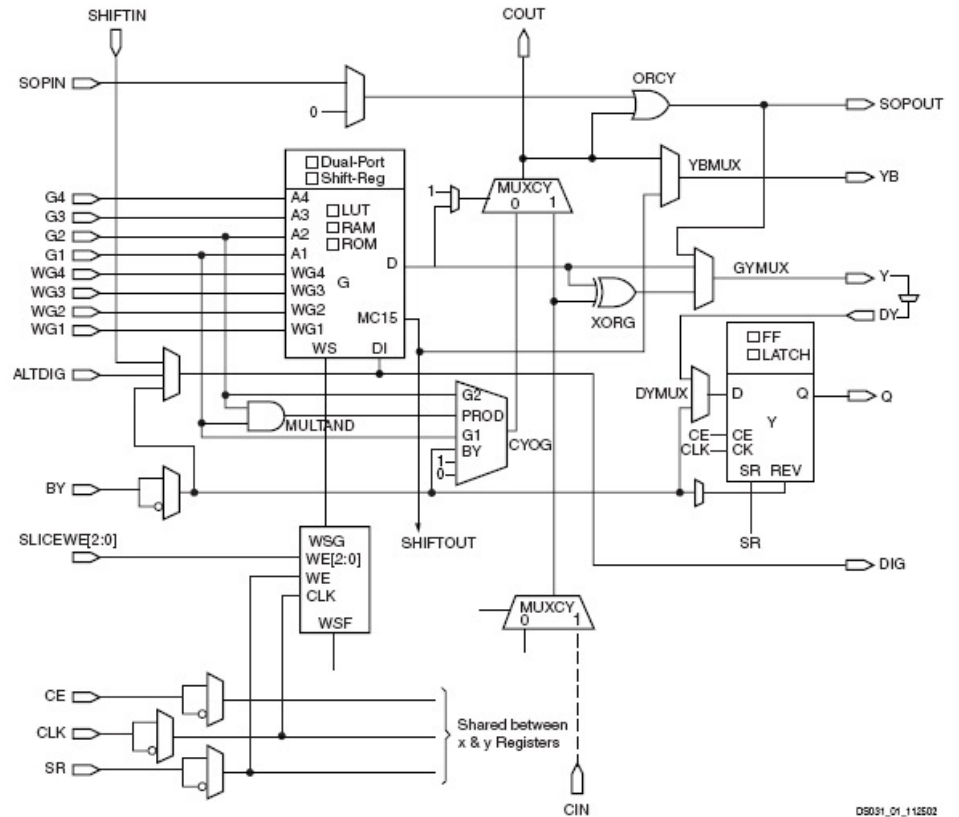


Digitally Controlled Impedance (DCI)

A practical example: Xilinx Virtex II Pro family



Slice

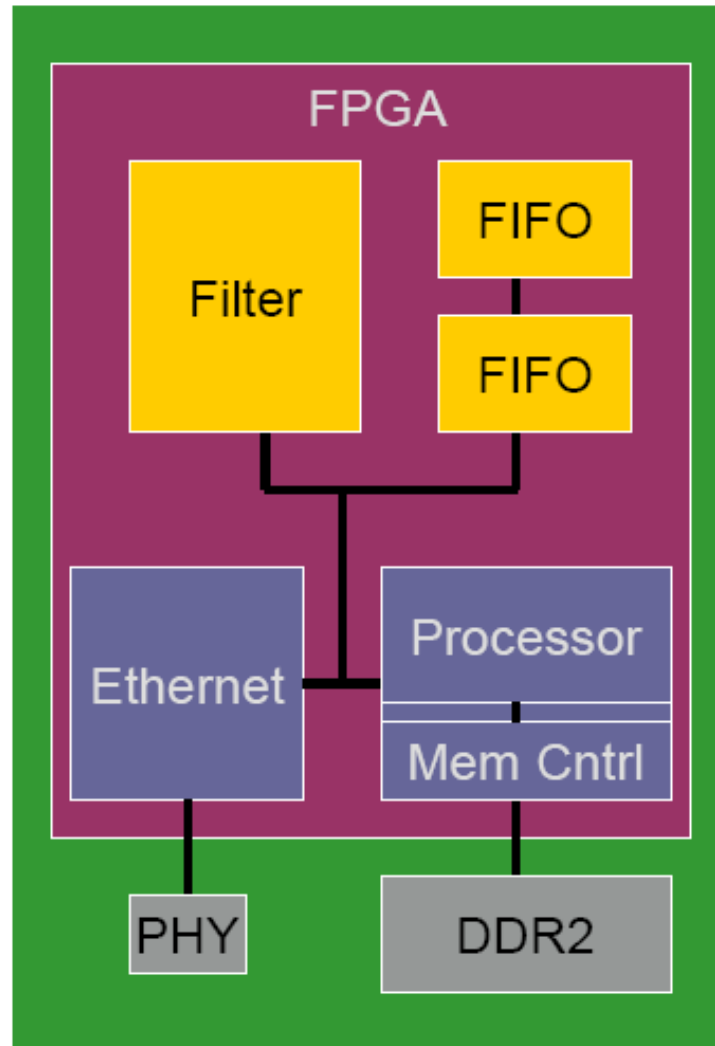


Detail of half-slice

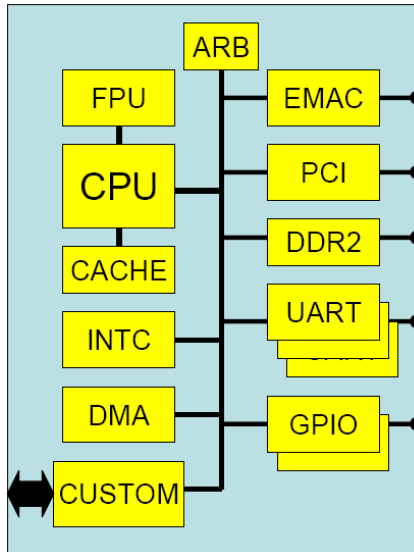
FPGA state of the art

- In addition to logic gates and routing, in a modern FPGA you can find:
 - Embedded processors (soft or hard).
 - Multi-Gb/s transceivers with equalization and hard IP for serial standards as PCI Express and Gbit Ethernet.
 - Lots of embedded MAC units, with enough bits to implement single precision floating point arithmetic efficiently.
 - Lots of dual-port RAM.
 - Sophisticated clock management through DLLs and PLLs.
 - System monitoring infrastructure including ADCs.
 - On-substrate decoupling capacitors to ease PCB design.
 - Digitally Controlled Impedance to eliminate on-board termination resistors.

Embedded processors

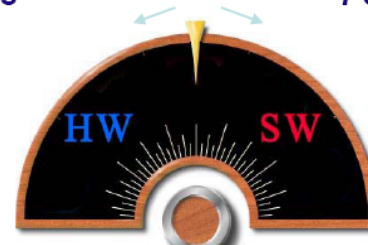
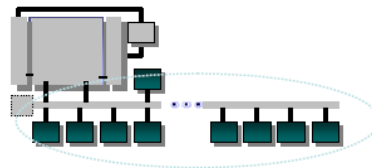


Why use embedded processors?

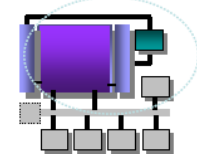


Customization: take only the peripherals you need and replicate them as many times as needed. Create your own custom peripherals.

Performing some software tasks in hardware can be expensive



Performing some hardware tasks in software can be slow



Strike optimum balance in system partitioning.

Supplementary discussion

Security Considerations

Yes, terrorists could have hacked Dick Cheney's heart

 By [Andrea Peterson](#)
October 21, 2013 at 8:58 a.m. EDT



The late [Barnaby Jack](#) was on the forefront of research into the vulnerability of medical devices. Last year, he demonstrated how a certain model of **implanted insulin pump** could be [lethally hacked](#) to administer incorrect dosages from up to 300 feet away. And this year he was expected to show off how implanted pacemakers and defibrillators could be hacked to deliver deadly shocks -- the almost exact type of issue Cheney's doctors were working to avoid. (From Washington Post)

Semiconductor supply chain

TSMC's Morris Chang receives SPIE Visionary Award for semiconductor industry leadership

Award recognizes 'revolutionary' changes, 'invention of the semiconductor foundry concept'

04 December 2014



Morris Chang (left), Chairman of TSMC, accepts the SPIE Visionary Award from SPIE CEO Eugene Arthurs during the OPTIC 2014 opening session. Organizers also kicked off the [International Year of Light \(IYL\) 2015](#) during a lighting ceremony that included LED-studded displays such as the IYL logo in the background.

TAICHUNG, TAIWAN and BELLINGHAM, Washington, USA -- Industry leader **Morris Chang**, Chairman of TSMC, was honored for his "invention of the semiconductor foundry concept" and other contributions with the rare presentation of an SPIE Visionary Award today (4 December) in Taichung. The award was made by **Eugene Arthurs**, CEO of SPIE, the international society for optics and photonics, during opening ceremonies of OPTIC (Optics and Photonics Taiwan, the International Conference) 2014. The event runs 4-5 December, along with the jointly organized annual meeting of the Taiwan Photonics Society.

